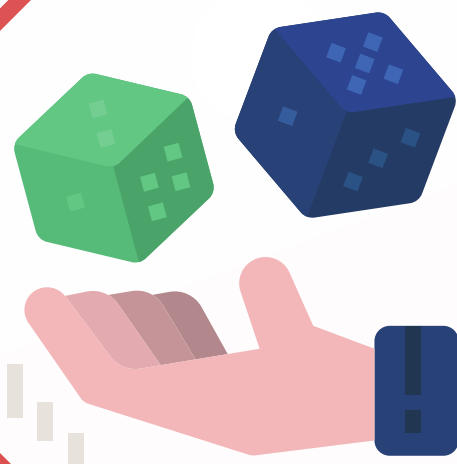




JEE (MAIN+ADVANCED)

Mathematics

ENTHUSIAST COURSE



STUDY MATERIAL

Probability

English Medium

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PROBABILITY

1 INTRODUCTION :

The theory of probability has been originated from the game of chance and gambling. In old days, gamblers used to gamble in a gambling house with a die to win the amount fixed among themselves. They were always desirous to get the prescribed number on the upper face of a die when it was thrown on a board. Shakuni of Mahabharat was perhaps one of them. People started to study the subject of probability from the middle of seventeenth century. The mathematicians Huygens, Pascal, Fermat and Bernoulli contributed a lot to this branch of Mathematics. A.N. Kolmogorow proposed the set theoretic model to the theory of probability.

Probability gives us a measure of likelihood that something will happen. However probability can never predict the number of times that an occurrence actually happens. But being able to quantify the likely occurrence of an event is important because most of the decisions that affect our daily lives are based on likelihoods and not on absolute certainties.

2. DEFINITIONS :

- (a) **Experiment** : An action or operation resulting in two or more well defined outcomes. e.g. tossing a coin, throwing a die, drawing a card from a pack of well shuffled playing cards etc.
- (b) **Sample space** : A set S that consists of all possible outcomes of a random experiment is called a sample space and each outcome is called a sample point. Often, there will be more than one sample space that can describe outcomes of an experiment, but there is usually only one that will provide the most information.

e.g. in an experiment of "throwing a die", following sample spaces are possible :

- (i) {even number, odd number}
- (ii) {a number less than 3, a number equal to 3, a number greater than 3}
- (iii) {1,2,3,4,5,6}

Here 3rd sample space is the one which provides most information.

If a sample space has a finite number of points it is called finite sample space and if it has an infinite number of points, it is called infinite sample space. e.g. (i) "in a toss of coin" either a head (H) or tail (T) comes up, therefore sample space of this experiment is $S = \{H, T\}$ which is a finite sample space. (ii) "Selecting a number from the set of natural numbers", sample space of this experiment is $S = \{1, 2, 3, 4, \dots\}$ which is an infinite sample space.

- (c) **Event** : An event is defined as an occurrence or situation, for example
- (i) in a toss of a coin, it shows head,
 - (ii) scoring a six on the throw of a die,
 - (iii) winning the first prize in a raffle,
 - (iv) being dealt a hand of four cards which are all clubs.

In every case it is set of some or all possible outcomes of the experiment. Therefore event (A) is subset of sample space (S). If outcome of an experiment is an element of A we say that event A has occurred.

- An event consisting of a single point of S is called a simple or elementary event.
- ϕ is called impossible event and S (sample space) is called sure event.

Note : Probability of occurrence of an event A is denoted by $P(A)$.

- (d) **Compound Event :** If an event has more than one sample points it is called **Compound Event**. If A & B are two given events then $A \cap B$ is called compound event and is denoted by $A \cap B$ or AB or A & B.
- (e) **Complement of an event :** The set of all outcomes which are in S but not in A is called the complement of the event A & denoted by $\bar{A}$, A^c , A' or 'not A'.
- (f) **Mutually Exclusive Events :** Two events are said to be **Mutually Exclusive** (or disjoint or incompatible) if the occurrence of one precludes (rules out) the simultaneous occurrence of the other. If A & B are two mutually exclusive events then $P(A \cap B) = 0$.

Consider, for example, choosing numbers at random from the set $\{3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

If, Event A is the selection of a prime number,

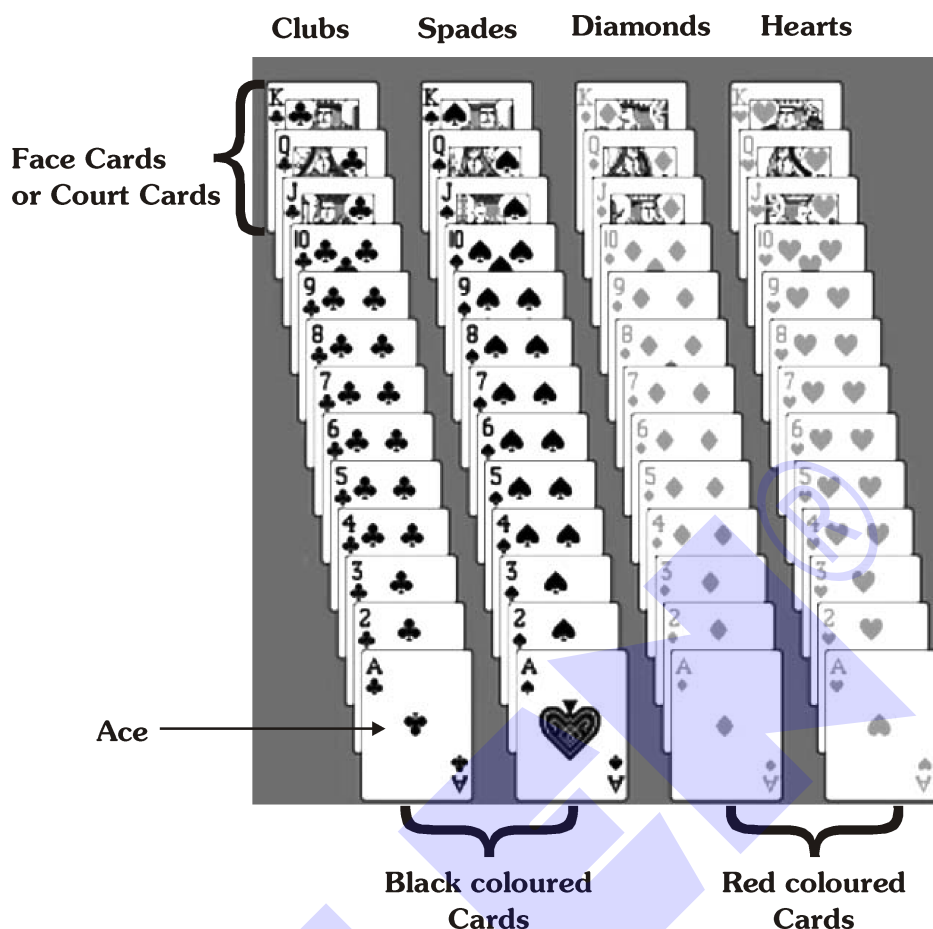
Event B is the selection of an odd number,

Event C is the selection of an even number,

then A and C are mutually exclusive as none of the numbers in this set is both prime and even. But A and B are not mutually exclusive as some numbers are both prime and odd (viz. 3, 5, 7, 11).

- (g) **Equally Likely Events :** Events are said to be **Equally Likely** when each event is as likely to occur as any other event. Note that the term 'at random' or 'randomly' means that all possibilities are equally likely.
- (h) **Exhaustive Events :** Events A, B, C, N are said to be **Exhaustive Events** if no event outside this set can result as an outcome of an experiment. For example, if A & B are two events defined on a sample space S and A & B are exhaustive $\Rightarrow A \cup B = S \Rightarrow P(A \cup B) = 1$.

Note : Playing cards : A pack of playing cards consists of 52 cards of 4 suits, 13 in each, as shown in figure.



Comparative study of Equally likely, Mutually Exclusive and Exhaustive events :

Experiment	Events	E/L	M/E	Exhaustive
1. Throwing of a die	A: throwing an odd face {1, 3, 5} B : throwing a composite {4,6}	No	Yes	No
2. A ball is drawn from an urn containing 2White, 3Red and 4Green balls	E_1 : getting a White ball E_2 : getting a Red ball E_3 : getting a Green ball	No	Yes	Yes
3. Throwing a pair of dice	A : throwing a doublet {11, 22, 33, 44, 55, 66} B : throwing a total of 10 or more { 46, 64, 55, 56, 65, 66 }	Yes	No	No
4. From a well shuffled pack of cards a card is drawn	E_1 : getting a heart E_2 : getting a spade E_3 : getting a diamond E_4 : getting a club	Yes	Yes	Yes
5. From a well shuffled pack of cards a card is drawn	A = getting a heart B = getting a face card	No	No	No

Illustration 1 : A coin is tossed. If it shows head, we draw a ball from a bag consisting of 3 blue and 4 white balls; if it shows tail we throw a die. Describe the sample space of this experiment.

Solution : Let us denote blue balls by B_1, B_2, B_3 and the white balls by W_1, W_2, W_3, W_4 . Then a sample space of the experiment is

$$S = \{HB_1, HB_2, HB_3, HW_1, HW_2, HW_3, HW_4, T1, T2, T3, T4, T5, T6\}.$$

Here HB_i means head on the coin and ball B_i is drawn, HW_i means head on the coin and ball W_i is drawn. Similarly, T_i means tail on the coin and the number i on the die.

Illustration 2 : Consider the experiment in which a coin is tossed repeatedly until a head comes up. Describe the sample space.

Solution : In the experiment head may come up on the first toss, or the 2nd toss, or the 3rd toss and so on. Hence, the desired sample space is $S = \{H, TH, TTH, TTTH, TTTTH, \dots\}$

Illustration 3 : A coin is tossed three times, consider the following events.

A : 'no head appears'

B : 'exactly one head appears'

C : 'at least two heads appear'

Do they form a set of mutually exclusive and exhaustive events ?

Solution : The sample space of the experiment is

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

Events A, B and C are given by

$$A = \{TTT\}$$

$$B = \{HTT, THT, TTH\}$$

$$C = \{HHT, HTH, THH, HHH\}$$

Now,

$$A \cup B \cup C = \{TTT, HTT, THT, TTH, HHT, HTH, THH, HHH\} = S$$

Therefore A, B and C are exhaustive events. Also, $A \cap B = \phi$, $A \cap C = \phi$ and $B \cap C = \phi$.

Therefore, the events are pair-wise disjoint, i.e., they are mutually exclusive. Hence, A, B and C form a set of mutually exclusive and exhaustive events.

Do yourself - 1 :

- Two balls are drawn from a bag containing 2 Red and 3 Black balls, write sample space of this experiment.
- Out of 2 men and 3 women a team of two persons is to be formed such that there is exactly one man and one woman. Write the sample space of this experiment.
- A coin is tossed and if head comes up, a die is thrown. But if tail comes up, the coin is tossed again. Write the sample space of this experiment.
- In a toss of a die, consider following events :

A : An even number turns up. B : A prime number turns up.

These events are -

(A) Equally likely events

(B) Mutually exclusive events

(C) Exhaustive events

(D) None of these

3. CLASSICAL DEFINITION OF PROBABILITY :

If n represents the total number of equally likely, mutually exclusive and exhaustive outcomes of an experiment and m of them are favourable to the happening of the event A , then the probability of happening of the event A is given by $P(A) = m/n$. There are $(n-m)$ outcomes which are favourable to the event that A does not happen. 'The event A does not happen' is denoted by $\bar{A}$ (and is read as 'not A ')

$$\text{Thus } P(\bar{A}) = \frac{n-m}{n} = 1 - \frac{m}{n}$$

$$\text{i.e. } P(\bar{A}) = 1 - P(A)$$

Note :

$$(i) \quad 0 \leq P(A) \leq 1$$

$$(ii) \quad P(A) + P(\bar{A}) = 1,$$

$$(iii) \quad \text{If } x \text{ cases are favourable to } A \text{ \& } y \text{ cases are favourable to } \bar{A} \text{ then } P(A) = \frac{x}{(x+y)} \text{ and}$$

$$P(\bar{A}) = \frac{y}{(x+y)}. \text{ We say that Odds In Favour Of } A \text{ are } x : y \text{ \& Odds Against } A \text{ are } y : x$$

OTHER DEFINITIONS OF PROBABILITY :

(a) Axiomatic probability : Axiomatic approach is another way of describing probability of an event. In this approach some axioms or rules are depicted to assign probabilities.

Let S be the sample space of a random experiment. The probability P is a real valued function whose domain is the power set of S and range is the interval $[0, 1]$ satisfying the following axioms :

$$(i) \quad \text{For any event } E, P(E) \geq 0$$

$$(ii) \quad P(S) = 1$$

$$(iii) \quad \text{If } E \text{ and } F \text{ are mutually exclusive events, then } P(E \cup F) = P(E) + P(F).$$

It follows from (iii) that $P(E \cap F) = P(\phi) = 0$.

Let S be a sample space containing outcomes $\omega_1, \omega_2, \dots, \omega_n$, i.e., $S = \{\omega_1, \omega_2, \dots, \omega_n\}$

It follows from the axiomatic definition of probability that :

$$(i) \quad 0 \leq P(\omega_i) \leq 1 \text{ for each } \omega_i \in S$$

$$(ii) \quad P(\omega_1) + P(\omega_2) + \dots + P(\omega_n) = 1$$

$$(iii) \quad \text{For any event } A, P(A) = \sum P(\omega_i), \omega_i \in A.$$

(b) Empirical probability : The probability that you would hit the bull's-eye on a dartboard with one throw of a dart would depend on how much you had practised, how much natural talent for playing darts you had, how tired you were, how good a dart you were using etc. all of which are impossible to quantify. A method which can be adopted in the example given above is to throw the dart several times (each throw is a trial) and count the number of times you hit the bull's-eye (a success) and the number of times you miss (a failure). Then an empirical value of the probability

$$\text{that you hit the bull's-eye with any one throw is } \frac{\text{number of successes}}{\text{number of successes} + \text{number of failures}}.$$

If the number of throws is small, this does not give a particular good estimate but for a large number of throws the result is more reliable.

When the probability of the occurrence of an event A cannot be worked out exactly, an empirical value can be found by adopting the approach described above, that is :

- (i) making a large number of trials (i.e. set up an experiment in which the event may, or may not, occur and note the outcome),
- (ii) counting the number of times the event does occur, i.e. the number of successes,
- (iii) calculating the value of $\frac{\text{number of successes}}{\text{number of trials (i.e. successes + failures)}} = \frac{r}{n}$

The probability of event A occurring is defined as $P(A) = \lim_{n \rightarrow \infty} \left(\frac{r}{n} \right)$

$n \rightarrow \infty$ means that the number of trials is large (but what should be taken as 'large' depends on the problem).

Illustration 4 : If the letters of INTERMEDIATE are arranged, then the odds in favour of the event that no two 'E's occur together, are-

- (A) $\frac{6}{5}$ (B) $\frac{5}{6}$ (C) $\frac{2}{9}$ (D) none of these

Solution : I $\rightarrow$ 2, N $\rightarrow$ 1, T $\rightarrow$ 2, E $\rightarrow$ 3, R $\rightarrow$ 1, M $\rightarrow$ 1, D $\rightarrow$ 1, A $\rightarrow$ 1 (3'E's, Rest 9 letters)

First arrange rest of the letters = $\frac{9!}{2! \ 2!}$,

Now 3'E's can be placed by ${}^{10}C_3$ ways, so favourable cases = $\frac{9!}{2! \ 2!} \times {}^{10}C_3 = 3 \times 10!$

Total cases = $\frac{12!}{2! \ 2! \ 3!} = \frac{11}{2} \times 10!$; Non-favourable cases = $\left(\frac{11}{2} - 3 \right) \times 10! = \frac{5}{2} \times 10!$

Odds in favour of the event = $\frac{3}{5/2} = \frac{6}{5}$

Ans. (A)

Illustration 5 : From a group of 10 persons consisting of 5 lawyers, 3 doctors and 2 engineers, four persons are selected at random. The probability that the selection contains at least one of each category is-

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) none of these

Solution : $n(S) = {}^{10}C_4 = 210$.

$n(E) = {}^5C_2 \times {}^3C_1 \times {}^2C_1 + {}^5C_1 \times {}^3C_2 \times {}^2C_1 + {}^5C_1 \times {}^3C_1 \times {}^2C_2 = 105$

$\therefore P(E) = \frac{105}{210} = \frac{1}{2}$.

Ans. (A)

Illustration 6 : If four cards are drawn at random from a pack of fifty-two playing cards, find the probability that at least one of them is an ace.

Solution : If A is a combination of four cards containing at least one ace (i.e. either one ace, or two aces, or three aces or four aces) then $\bar{A}$ is a combination of four cards containing no aces.

$$\text{Now } P(\bar{A}) = \frac{\text{Number of combinations of four cards with no aces}}{\text{Total number of combinations of four cards}} = \frac{{}^{48}C_4}{{}^{52}C_4} = 0.72$$

$$\text{Using } P(A) + P(\bar{A}) = 1 \text{ we have } P(A) = 1 - P(\bar{A}) = 1 - 0.72 = 0.28$$

Illustration 7 : A bag contains n white and n red balls. Pairs of balls are drawn without replacement until the bag is empty. Show that the probability that each pair consists of one white and one red ball is $2^n / (2^n C_n)$.

Solution : Let S be the sample space & E be the event that each of the n pairs of balls drawn consists of one white and one red ball.

$$\begin{aligned} \therefore n(S) &= ({}^{2n}C_2) ({}^{2n-2}C_2) ({}^{2n-4}C_2) \dots ({}^4C_2) ({}^2C_2) \\ &= \left\{ \frac{(2n)(2n-1)}{1.2} \right\} \left\{ \frac{(2n-2)(2n-3)}{1.2} \right\} \left\{ \frac{(2n-4)(2n-5)}{1.2} \right\} \dots \left\{ \frac{4.3}{1.2} \right\} \left\{ \frac{2.1}{1.2} \right\} \\ &= \frac{1.2.3.4 \dots (2n-1)2n}{2^n} = \frac{2n!}{2^n} \end{aligned}$$

$$\begin{aligned} \text{and } n(E) &= ({}^nC_1 \cdot {}^nC_1) ({}^{n-1}C_1 \cdot {}^{n-1}C_1) ({}^{n-2}C_1 \cdot {}^{n-2}C_1) \dots ({}^2C_1 \cdot {}^2C_1) ({}^1C_1 \cdot {}^1C_1) \\ &= n^2 \cdot (n-1)^2 \cdot (n-2)^2 \dots 2^2 \cdot 1^2 = [1.2.3 \dots (n-1)n]^2 = (n!)^2 \end{aligned}$$

$$\therefore \text{ Required Probability, } P(E) = \frac{n(E)}{n(S)} = \frac{(n!)^2}{(2n)!/2^n} = \frac{2^n}{2n!} = \frac{2^n}{(n!)^2} \quad \text{Ans.}$$

Do yourself - 2 :

1. A coin is tossed successively three times. Find the probability of getting exactly one head or two heads.
2. A bag contains 5 red and 4 green balls. Four balls are drawn at random then find the probability that two balls are of red and two balls are of green colour.
3. Two natural numbers are selected at random, find the probability that their sum is divisible by 10.
4. Five card are drawn successively from a pack of 52 cards with replacement. Find the probability that there is at least one Ace.
5. A cube with all six faces coloured is cut into 64 cubical blocks of the same size which are thoroughly mixed. Find the probability that the 2 randomly chosen blocks have 2 coloured faces each -

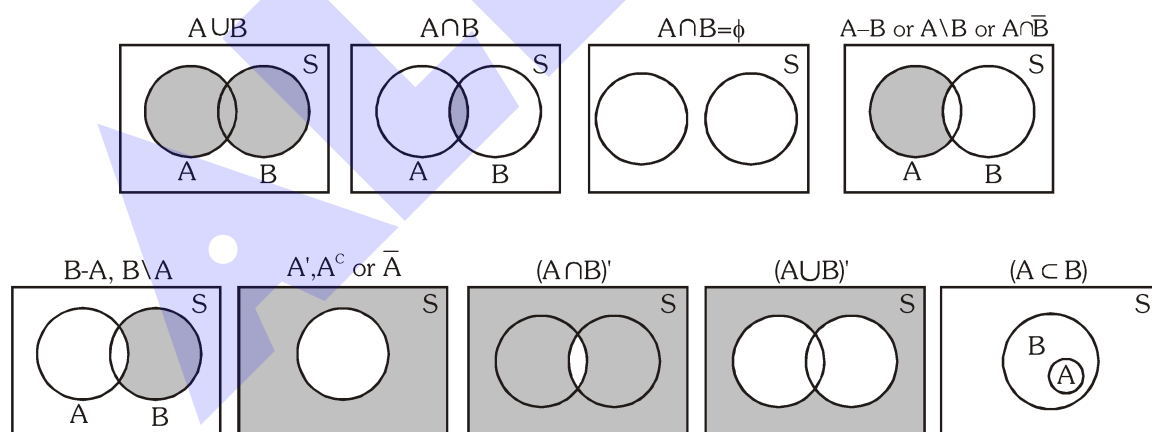
6. A card is drawn at random from a well shuffled deck of cards. Find the probability that the card is a-
- (i) king or a red card (ii) club or a diamond (iii) king or a queen
(iv) king or an ace (v) spade or a club (vi) neither a heart nor a king
7. A bag contain 5 white, 7 black, and 4 red balls, find the chance that three balls drawn at random are all white.
8. Thirteen persons take their places at a round table, Find the odds against two particular persons sitting together.
9. Consider a function $f(x)$ that has zeroes 4 and 9. Given that Mr. A randomly selects a number from the set $\{-10, -9, -8, \dots, 8, 9, 10\}$, what is the probability that Mr. A chooses a zero of $f(x^2)$?
10. (a) A fair die is tossed. If the number is odd, find the probability that it is prime.
(b) Three fair coins are tossed. If both heads and tails appear, determine the probability that exactly one head appears.
11. Mr. A lives at origin on the cartesian plane and has his office at (4,5). His friend lives at (2,3) on the same plane. Mr. A can go to his office travelling one block at a time either in the +y or +x direction. If all possible paths are equally likely then the probability that Mr. A passed his friends house is -
(A) $1/2$ (B) $10/21$ (C) $1/4$ (D) $11/21$
12. I have 3 normal dice, one red, one blue and one green and I roll all three simultaneously. Let P be the probability that the sum of the numbers on the red and blue dice is equal to the number on the green die. If P is the written in lowest terms as a/b then the value of $(a + b)$ equals -
(A) 79 (B) 77 (C) 61 (D) 57
13. There are three passengers on an airport shuttle bus that makes stops at four different hotels. The probability that all three passengers will be staying at different hotels, is -
(A) $\frac{1}{16}$ (B) $\frac{1}{4}$ (C) $\frac{3}{8}$ (D) $\frac{3}{4}$
14. n different books ($n \geq 3$) are put at random in a shelf. Among these books there is a particular book 'A' and a particular book B. The probability that there are exactly 'r' books between A and B is -
(A) $\frac{2}{n(n-1)}$ (B) $\frac{2(n-r-1)}{n(n-1)}$ (C) $\frac{2(n-r-2)}{n(n-1)}$ (D) $\frac{(n-r)}{n(n-1)}$
15. Of all the mappings that can be defined from the set $A : \{1, 2, 3, 4\} \rightarrow B(5, 6, 7, 8, 9)$, a mapping is randomly selected. The chance that the selected mapping is strictly monotonic, is
(A) $\frac{1}{125}$ (B) $\frac{2}{125}$ (C) $\frac{5}{4096}$ (D) $\frac{5}{2048}$

16. A coin is biased so that heads is three times as likely to appear as tails. Find $P(H)$ and $P(T)$. If such a coin is tossed twice find the probability that head occurs at least once.
17. In a single throw of three dice, determine the probability of getting
(i) a total of 5 (ii) a total of atmost 5 (iii) a total of at least 5.
18. A natural number x is randomly selected from the set of first 100 natural numbers. Find the probability that it satisfies the inequality. $x + \frac{100}{x} > 50$
19. 3 students A, B and C are in a swimming race. A and B have the same probability of winning and each is twice as likely to win as C. Find the probability that B or C wins. Assume no two reach the winning point simultaneously.
20. A box contains 7 tickets, numbered from 1 to 7 inclusive. If 3 tickets are drawn from the box without replacement, one at a time, determine the probability that they are alternatively either odd-even-odd or even-odd-even.
21. 5 different marbles are placed in 5 different boxes randomly. Find the probability that exactly two boxes remain empty. Given each box can hold any number of marbles.

4. VENN DIAGRAMS :

A diagram used to illustrate relationships between sets. Commonly, a rectangle represents the universal set and a circle within it represents a given set (all members of the given set are represented by points within the circle). A subset is represented by a circle within a circle and intersection is indicated by overlapping circles.

Let S is the sample space of an experiment and A, B are two events corresponding to it :

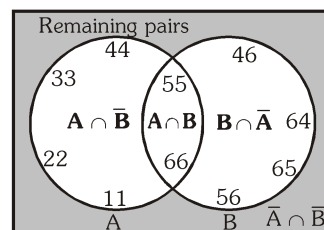


Example : Let us conduct an experiment of tossing a pair of dice.

Two events defined on the experiment are

A : getting a doublet { 11, 22, 33, 44, 55, 66 }

B : getting total score of 10 or more { 64, 46, 55, 56, 65, 66 }

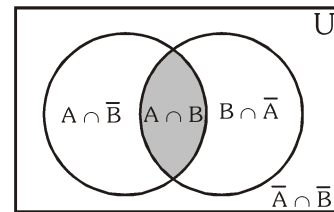


5. ADDITION THEOREM :

$A \cup B = A + B = A \text{ or } B$ denotes occurrence of at least A or B.

For 2 events A & B :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



$$\begin{aligned}
 & \left. \begin{aligned}
 & \text{(a) } P(A \cup B) \\
 & P(A + B) \\
 & P(A \text{ or } B) \\
 & P(\text{occurrence of at least A or B})
 \end{aligned} \right\} = \begin{aligned}
 & P(A) + P(B) - P(A \cap B) \text{ (This is known as generalised addition theorem)} \\
 & P(A) + P(B \cap \bar{A}) \\
 & P(B) + P(A \cap \bar{B}) \\
 & P(A \cap \bar{B}) + P(A \cap B) + P(B \cap \bar{A}) \\
 & 1 - P(\bar{A} \cap \bar{B}) \\
 & 1 - P(\overline{A \cup B})
 \end{aligned}
 \end{aligned}$$

Note :

- (i) If A & B are mutually exclusive then $P(A \cup B) = P(A) + P(B)$.
- (ii) If A & B are mutually exclusive and exhaustive, then $P(A \cup B) = P(A) + P(B) = 1$
- (b) $P(\text{only A occurs}) = P(A \setminus B) = P(A - B) = P(A \cap B^c) = P(A) - P(A \cap B)$
- (c) $P(\text{either A or B}) = 1 - P(\text{neither A nor B})$
i.e. $P(A \cup B) = 1 - P(\bar{A} \cap \bar{B})$
- (d) For any two events A & B
 $P(\text{exactly one of A, B occurs}) = P(A \cap \bar{B}) + P(B \cap \bar{A})$
 $\Rightarrow P(\text{exactly one of A, B occurs}) = P(A) + P(B) - 2P(A \cap B)$
 $= P(A \cup B) - P(A \cap B) = P(A^c \cup B^c) - P(A^c \cap B^c)$
- (e) $P(A \cap B) \leq P(A), P(B) \leq P(A \cup B) \leq P(A) + P(B)$

6. DE MORGAN'S LAW :

If A & B are two subsets of a universal set U, then

- (i) $(A \cup B)^c = A^c \cap B^c$ & (ii) $(A \cap B)^c = A^c \cup B^c$

Note :

- (a) $(A \cup B \cup C)^c = A^c \cap B^c \cap C^c$ & $(A \cap B \cap C)^c = A^c \cup B^c \cup C^c$
- (b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ & $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Illustration 8 : Given two events A and B. If odds against A are as 2 : 1 and those in favour of $A \cup B$ are as 3 : 1, then find the range of $P(B)$.

Solution : Clearly $P(A) = 1/3$, $P(A \cup B) = 3/4$.

$$\text{Now, } P(B) \leq P(A \cup B)$$

$$\Rightarrow P(B) \leq 3/4$$

$$\text{Also, } P(B) = P(A \cup B) - P(A) + P(A \cap B)$$

$$\Rightarrow P(B) \geq P(A \cup B) - P(A) \quad (\because P(A \cap B) \geq 0)$$

$$\Rightarrow P(B) \geq 3/4 - 1/3 \quad \Rightarrow P(B) \geq \frac{5}{12}$$

$$\Rightarrow \frac{5}{12} \leq P(B) \leq \frac{3}{4}$$

Ans.

Illustration 9 : If A and B are two events such that $P(A \cup B) = \frac{3}{4}$, $P(A \cap B) = \frac{1}{4}$ and $P(A^c) = \frac{2}{3}$. Then find -

(i) $P(A)$

(ii) $P(B)$

(iii) $P(A \cap B^c)$

(iv) $P(A^c \cap B)$

Solution : $P(A) = 1 - P(A^c) = 1 - \frac{2}{3} = \frac{1}{3}$

$$P(B) = P(A \cup B) + P(A \cap B) - P(A) = \frac{3}{4} + \frac{1}{4} - \frac{1}{3} = \frac{2}{3}$$

$$P(A \cap B^c) = P(A) - P(A \cap B) = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

$$P(A^c \cap B) = P(B) - P(A \cap B) = \frac{2}{3} - \frac{1}{4} = \frac{5}{12}$$

Ans.

Do yourself - 3 :

- Draw Venn diagram of (a) $(A^c \cap B^c) \cup (A \cap B)$ (b) $B^c \cup (A^c \cap B)$
- If A and B are two mutually exclusive events, then-
 (A) $P(A) \leq P(\bar{B})$ (B) $P(\bar{A} \cap \bar{B}) = P(\bar{A}) - P(B)$
 (C) $P(\bar{A} \cup \bar{B}) = 0$ (D) $P(\bar{A} \cap B) = P(B)$
- A bag contains 6 white, 5 black and 4 red balls. Find the probability of getting either a white or a black ball in a single draw.
- In a class of 125 students, 70 passed in English, 55 in mathematics and 30 in both. Find the probability that a student selected at random from the class has passed in
 (a) at least one subject (b) only one subject.
- In throwing 3 dice, the probability that atleast 2 of the three numbers obtained are same is -
 (A) $1/2$ (B) $1/3$ (C) $4/9$ (D) none

6. From a pack of 52 playing cards, face cards and tens are removed and kept aside then a card is drawn at random from the remaining cards. If
 A: The event that the card drawn is an ace
 H: The event that the card drawn is a heart
 S : The event that the card drawn is a spade
 then which of the following holds ?
 (A) $9P(A) = 4P(H)$ (B) $P(S) = 4P(A \cap H)$
 (C) $3P(H) = 4P(A \cup S)$ (D) $P(H) = 12P(A \cap S)$
7. Given two independent events A, B such that $P(A) = 0.3$, $P(B) = 0.6$. Determine
 (i) $P(A \text{ and } B)$ (ii) $P(A \text{ and not } B)$ (iii) $P(\text{not } A \text{ and } B)$
 (iv) $P(\text{neither } A \text{ nor } B)$ (v) $P(A \text{ or } B)$
8. The probabilities that a student will receive A, B, C or D grade are 0.40, 0.35, 0.15 and 0.10 respectively. Find the probability that a student will receive
 (i) not an A grade (ii) B or C grade (iii) at most C grade
9. Let a red die, a blue die, a green die and a white die are rolled once, the dice being fair. The outcomes on the red, blue, green and white die denote the numbers, a, b, c and d respectively. Let E denotes the event that absolute value of $(a - 1)(b - 2)(c - 3)(d - 6) = 1$, then $P(E)$ is -
 (A) $\frac{1}{324}$ (B) $\frac{1}{648}$ (C) $\frac{2}{324}$ (D) $\frac{1}{162}$
10. Let A and B be events such that $P(\bar{A}) = 4/5$, $P(B) = 1/3$, $P(A/B) = 1/6$, then :
 (a) $P(A \cap B)$ (b) $P(A \cup B)$
 (c) $P(B/A)$ (d) Are A and B independent?
11. In a given race, the odds in favour of four horses A, B, C & D are 1 : 3, 1 : 4, 1 : 5 and 1 : 6 respectively. Assuming that a dead heat is impossible, find the chance that one of them wins the race.
12. The probability that a person will get an electric contract is $2/5$ and the probability that he will not get plumbing contract is $4/7$. If the probability of getting at least one contract is $2/3$, what is the probability that he will get both?

7. CONDITIONAL PROBABILITY AND MULTIPLICATION THEOREM :

- (a) **Conditional Probability :** Let A and B be two events such that $P(A) > 0$. Then $P(B|A)$ denote the conditional probability of B given that A has occurred. Since A is known to have occurred, it becomes the new sample space replacing the original S. From this we led to the definition

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \text{which is called conditional probability of B given A}$$

- (b) **Multiplication Theorem :** $P(A \cap B) = P(A) P(B|A)$ which is called compound probability or multiplication theorem. It says the probability that both A and B occur is equal to the probability that A occur times the probability that B occurs given that A has occurred.

Note : For any three events A_1, A_2, A_3 we have

$$P(A_1 \cap A_2 \cap A_3) = P(A_1) P(A_2|A_1) P(A_3|(A_1 \cap A_2))$$

Illustration 10 : Two dice are thrown. Find the probability that the numbers appeared have a sum of 8 if it is known that the second die always exhibits 4

Solution : Let A be the event of occurrence of 4 always on the second die

$$= \{(1, 4), (2, 4), (3, 4), (4, 4), (5, 4), (6, 4)\} ; \quad \therefore n(A) = 6$$

and B be the event of occurrence of such numbers on both dice whose sum is 8 = $\{(6, 2), (5, 3), (4, 4), (3, 5), (2, 6)\}$.

$$\text{Thus, } A \cap B = A \cap \{(4, 4)\} = \{(4, 4)\}$$

$$\therefore n(A \cap B) = 1$$

$$\therefore P(B/A) = \frac{n(A \cap B)}{n(A)} = \frac{1}{6} \quad \text{or} \quad \frac{P(A \cap B)}{P(A)} = \frac{1/36}{6/36} = \frac{1}{6}$$

Illustration 11 : A bag contains 3 red, 6 white and 7 blue balls. Two balls are drawn one by one. What is the probability that first ball is white and second ball is blue when first drawn ball is not replaced in the bag?

Solution : Let A be the event of drawing first ball white and B be the event of drawing second ball blue.

Here A and B are dependent events.

$$P(A) = \frac{6}{16}, \quad P(B|A) = \frac{7}{15}$$

$$P(AB) = P(A) \cdot P(B|A) = \frac{6}{16} \times \frac{7}{15} = \frac{7}{40}$$

Illustration 12 : A bag contains 4 red and 4 blue balls. Four balls are drawn one by one from the bag, then find the probability that the drawn balls are in alternate colour.

Solution : E_1 : Event that first drawn ball is red, second is blue and so on.

E_2 : Event that first drawn ball is blue, second is red and so on.

$$\therefore P(E_1) = \frac{4}{8} \times \frac{4}{7} \times \frac{3}{6} \times \frac{3}{5} \quad \text{and} \quad P(E_2) = \frac{4}{8} \times \frac{4}{7} \times \frac{3}{6} \times \frac{3}{5}$$

$$P(E) = P(E_1) + P(E_2) = 2 \times \frac{4}{8} \cdot \frac{4}{7} \cdot \frac{3}{6} \cdot \frac{3}{5} = \frac{6}{35}$$

Ans.

Illustration 13 : If two events A and B are such that $P(\bar{A}) = 0.3$, $P(B) = 0.4$ and $P(A\bar{B}) = 0.5$ then

$P(B|(A \cup \bar{B}))$ equals -

- (A) $1/2$ (B) $1/3$ (C) $1/4$ (D) $1/5$

Solution : We have $P(B|(A \cup \bar{B})) = \frac{P[B \cap (A \cup \bar{B})]}{P(A \cup \bar{B})} = \frac{P[(B \cap A) \cup (B \cap \bar{B})]}{P(A) + P(\bar{B}) - P(A \cap \bar{B})}$

$$= \frac{P(AB)}{P(A) + P(\bar{B}) - P(A\bar{B})} = \frac{P(A) - P(A\bar{B})}{P(A) + P(\bar{B}) - P(A\bar{B})} = \frac{0.7 - 0.5}{0.7 + 0.6 - 0.5} = \frac{0.2}{0.8} = \frac{1}{4} \text{ Ans. (C)}$$

Illustration 14 : Three coins are tossed. Two of them are fair and one is biased so that a head is three times as likely as a tail. Find the probability of getting two heads and a tail.

Solution : E_1 : Event that head occurs on fair coin

E_2 : Event that, head occurs on biased coin

$$\therefore P(E_1) = \frac{1}{2}, P(E_2) = \frac{3}{4}$$

E : HHT or HTH or THH

$$\begin{aligned} \Rightarrow P(E) &= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{4} + \frac{1}{2} \times \frac{1}{2} \times \frac{3}{4} + \frac{1}{2} \times \frac{1}{2} \times \frac{3}{4} \\ &= \frac{7}{16} \end{aligned}$$

Illustration 15 : In a multiple choice test of three questions there are five alternative answers given to the first two questions each and four alternative answers given to the last question. If a candidate guesses answers at random, what is the probability that he will get-

- (a) Exactly one right answer ? (b) At least one right answer ?

Solution : E_1 : Event that, candidate guesses a correct answer for I question

E_2 : Event that, candidate guesses a correct answer for II question

E_3 : Event that, candidate guesses a correct answer for III question

$$\therefore P(E_1) = \frac{1}{5}, P(E_2) = \frac{1}{5}, P(E_3) = \frac{1}{4}$$

(a) E : Event that candidate get exactly one correct answer.

$$\therefore P(E) = P(E_1)P(\bar{E}_2)P(\bar{E}_3) + P(\bar{E}_1)P(E_2)P(\bar{E}_3) + P(\bar{E}_1)P(\bar{E}_2)P(E_3)$$

$$= \frac{1}{5} \cdot \frac{4}{5} \cdot \frac{3}{4} + \frac{4}{5} \cdot \frac{1}{5} \cdot \frac{3}{4} + \frac{4}{5} \cdot \frac{4}{5} \cdot \frac{1}{4} = \frac{2}{5}$$

(b) E : Event that candidate gets atleast one correct answer

$$\therefore P(E) = 1 - P(\bar{E}_1)P(\bar{E}_2)P(\bar{E}_3) = 1 - \frac{4}{5} \cdot \frac{4}{5} \cdot \frac{3}{4} = \frac{13}{25}$$

Illustration 16 : A speaks truth in 75% cases and B in 80% cases. What is the probability that they contradict each other in stating the same fact?

- (A) 7/20 (B) 13/20 (C) 3/20 (D) 1/5

Solution : There are two mutually exclusive cases in which they contradict each other i.e. $\bar{A}B$ and $A\bar{B}$. Hence required probability = $P(\bar{A}B + A\bar{B}) = P(\bar{A}B) + P(A\bar{B})$

$$= P(A)P(\bar{B}) + P(\bar{A})P(B) = \frac{3}{4} \cdot \frac{1}{5} + \frac{1}{4} \cdot \frac{4}{5} = \frac{7}{20} \quad \text{Ans. (A)}$$

Do yourself - 4 :

1. A bag contains 2 black, 4 white and 3 red balls. One ball is drawn at random from the bag and kept aside. From the remaining balls another ball is drawn and kept aside the first. This process is repeated till all the balls are drawn. Then probability that the balls drawn are in sequence of 2 black, 4 white and 3 red is-

- (A) $\frac{1}{1260}$ (B) $\frac{1}{7560}$ (C) $\frac{1}{210}$ (D) None of these

2. Three cards are drawn successively, without replacement from a pack of 52 well shuffled cards. What is the probability that the drawn cards are face cards of same suit ?

3. A pair of numbers is picked up randomly (without replacement) from the set $\{1, 2, 3, 5, 7, 11, 12, 13, 17, 19\}$. The probability that the number 11 was picked given that the sum of the numbers was even, is nearly :

- (A) 0.1 (B) 0.125 (C) 0.24 (D) 0.18

4. For a biased die the probabilities for the different faces to turn up are given below :

Faces :	1	2	3	4	5	6
Probabilities :	0.10	0.32	0.21	0.15	0.05	0.17

The die is tossed & you are told that either face one or face two has turned up. Then the probability that it is face one is :

- (A) 1/6 (B) 1/10 (C) 5/49 (D) 5/21

5. The probability that an automobile will be stolen and found within one week is 0.0006. The probability that an automobile will be stolen is 0.0015. The probability that a stolen automobile will be found in one week is

- (A) 0.3 (B) 0.4 (C) 0.5 (D) 0.6

6. A box contains 100 tickets numbered 1, 2, 3, ..., 100. Two tickets are chosen at random. It is given that the maximum number on the two chosen tickets is not more than 10. The minimum number on them is 5, with probability

- (A) $\frac{1}{9}$ (B) $\frac{2}{11}$ (C) $\frac{3}{19}$ (D) none

7. Two boys A and B find the jumble of n ropes lying on the floor. Each takes hold of one loose end randomly. If the probability that they are both holding the same rope is $\frac{1}{101}$ then the number of ropes is equal to
 (A) 101 (B) 100 (C) 51 (D) 50
8. A key to room number C_3 is dropped into a jar with five other keys, and the jar is thoroughly mixed. If keys are randomly drawn from the jar without replacement until the key to room C_3 is chosen, then what are the odds in favour that the key to room C_3 will be obtained on the 2nd try?
 (A) 1 : 4 (B) 1 : 5 (C) 1 : 6 (D) 5 : 6

8. INDEPENDENT EVENTS :

Two events A & B are said to be independent if occurrence or non occurrence of one does not affect the probability of the occurrence or non occurrence of other.

- (a) If the occurrence of one event affects the probability of the occurrence of the other event then the events are said to be **Dependent** or **Contingent**. For two independent events A and B : $P(A \cap B) = P(A) \cdot P(B)$. Often this is taken as the definition of independent events.

Note : If A and B are independent events, then

$$(i) P(\bar{A} \cap \bar{B}) = P(\bar{A}) \cdot P(\bar{B}) \quad (ii) P(A \cap \bar{B}) = P(A) \cdot P(\bar{B})$$

- (b) Three events A, B & C are independent if & only if all the following conditions hold ;

$$P(A \cap B) = P(A) \cdot P(B) ; \quad P(B \cap C) = P(B) \cdot P(C)$$

$$P(C \cap A) = P(C) \cdot P(A) \text{ and } P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$$

i.e. they must be pairwise as well as mutually independent.

- (c) If three events A, B and C are pair wise mutually exclusive then they must be mutually exclusive. i.e. $P(A \cap B) = P(B \cap C) = P(C \cap A) = 0 \Rightarrow P(A \cap B \cap C) = 0$. However the converse of this is not true.

Note :

Independent events are not in general mutually exclusive & vice versa. Mutually exclusiveness can be used when the events are taken from the same experiment & independence can be used when the events are taken from different experiments.

Illustration 17 : If A & B are independent events such that $P(A \cap \bar{B}) = \frac{1}{3}$ & $P(A \cup B) = \frac{11}{15}$,

then $P(A \cap B)$ is equal to

- (A) $\frac{1}{2}$ (B) $\frac{5}{11}$ (C) $\frac{2}{9}$ (D) $\frac{7}{9}$

Solution : $P(A) - P(A \cap B) = \frac{1}{3}$ & $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{11}{15}$

$$\Rightarrow P(B) = \frac{6}{15} = \frac{2}{5}$$

$$P(A) - P(A)P(B) = \frac{1}{3} \Rightarrow P(A) = \frac{5}{9}$$

$$\Rightarrow P(A \cap B) = P(A)P(B) = \frac{2}{5} \times \frac{5}{9} = \frac{2}{9}$$

Illustration 18 : A lot contains 50 defective and 50 non-defective bulbs. Two bulbs are drawn at random, one at a time, with replacement. The events A, B, C are defined as :

A = {The first bulb is defective}

B = {The second bulb is non-defective}

C = {The two bulbs are both defective or both non-defective}

Determine whether

(i) A, B, C are pairwise independent, (ii) A, B, C are independent.

Solution : We have $P(A) = \frac{50}{100} \cdot 1 = \frac{1}{2}$; $P(B) = 1 \cdot \frac{50}{100} = \frac{1}{2}$; $P(C) = \frac{50}{100} \cdot \frac{50}{100} + \frac{50}{100} \cdot \frac{50}{100} = \frac{1}{2}$

$A \cap B$ is the event that first bulb is defective and second is non-defective.

$$\therefore P(A \cap B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$A \cap C$ is the event that both bulbs are defective.

$$\therefore P(A \cap C) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$\text{Similarly } P(B \cap C) = \frac{1}{4}$$

Thus we have $P(A \cap B) = P(A) \cdot P(B)$; $P(A \cap C) = P(A) \cdot P(C)$; $P(B \cap C) = P(B) \cdot P(C)$

$\therefore$ A, B and C are pairwise independent.

There is no element in $A \cap B \cap C$

$$\therefore P(A \cap B \cap C) = 0$$

$$\therefore P(A \cap B \cap C) \neq P(A) \cdot P(B) \cdot P(C)$$

Hence A, B and C are not mutually independent.

Do yourself - 5 :

1. For two independent events A and B, the probability that both A & B occur is $\frac{1}{8}$ and the probability that neither of them occur is $\frac{3}{8}$. The probability of occurrence of A may be -
(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{3}{4}$
2. A die marked with numbers 1,2,2,3,3,3 is rolled three times. Find the probability of occurrence of 1,2 and 3 respectively.
3. Let A & B be two events. Suppose $P(A) = 0.4$, $P(B) = p$ & $P(A \cup B) = 0.7$. The value of p for which A & B are independent is :
(A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) $\frac{1}{5}$
4. Sixteen players $s_1, s_2, \dots, s_{16}$ play in a tournament. They are divided into eight pairs at random. From each pair a winner is decided on the basis of a game played between the two players of the pair. Assume that all the players are of equal strength. The probability that "exactly one of the two players s_1 & s_2 is among the eight winners" is
(A) $\frac{4}{15}$ (B) $\frac{7}{15}$ (C) $\frac{8}{15}$ (D) $\frac{9}{15}$
5. Events A and C are independent. If the probabilities relating A, B and C are $P(A) = \frac{1}{5}$; $P(B) = \frac{1}{6}$; $P(A \cap C) = \frac{1}{20}$; $P(B \cup C) = \frac{3}{8}$ then
(A) events B and C are independent
(B) events B and C are mutually exclusive
(C) events B and C are neither independent nor mutually exclusive
(D) events A and C are equiprobable
6. An unbiased cubic die marked with 1, 2, 2, 3, 3, 3 is rolled 3 times. The probability of getting a total score of 4 or 6 is
(A) $\frac{16}{216}$ (B) $\frac{50}{216}$ (C) $\frac{60}{216}$ (D) none
7. A biased coin with probability P, $0 < P < 1$, of heads, is tossed until a head appears for the first time. If the probability that the number of tosses required is even is $\frac{2}{5}$, then the value of P is
(A) $\frac{1}{4}$ (B) $\frac{1}{6}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$
8. When a missile is fired from a ship, the probability that it is intercepted is $\frac{1}{3}$. The probability that the missile hits the target, given that it is not intercepted is $\frac{3}{4}$. If three missiles are fired independently from the ship, the probability that all three hits the target, is
(A) $\frac{1}{12}$ (B) $\frac{1}{8}$ (C) $\frac{3}{8}$ (D) $\frac{3}{4}$

9. TOTAL PROBABILITY THEOREM :

Let an event A of an experiment occurs with its n mutually exclusive & exhaustive events $B_1, B_2, B_3, \dots, B_n$
then total probability of occurrence of even A is

$$P(A) = P(AB_1) + P(AB_2) + \dots + P(AB_n) = \sum_{i=1}^n P(AB_i)$$

$$P(A) = P(B_1) P(A|B_1) + P(B_2) P(A|B_2) + \dots + P(B_n) P(A|B_n) \\ = \sum P(B_i) P(A|B_i)$$

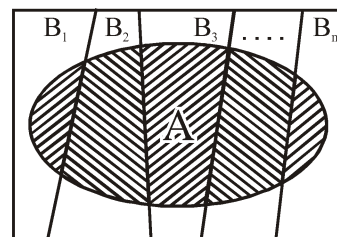


Illustration 19 : A purse contains 4 copper and 3 silver coins and another purse contains 6 copper and 2 silver coins. One coin is drawn from any one of these two purses. The probability that it is a copper coin is -

- (A) $\frac{4}{7}$ (B) $\frac{3}{4}$ (C) $\frac{2}{7}$ (D) $\frac{37}{56}$

Solution :

Let $A \equiv$ event of selecting first purse

$B \equiv$ event of selecting second purse

$C \equiv$ event of drawing a copper coin

Then given event has two disjoint cases: AC and BC

$$\therefore P(C) = P(AC + BC) = P(AC) + P(BC) = P(A)P(C|A) + P(B)P(C|B)$$

$$= \frac{1}{2} \cdot \frac{4}{7} + \frac{1}{2} \cdot \frac{6}{8} = \frac{37}{56}$$

Ans. (D)

Illustration 20 : Three groups A, B, C are contesting for positions on the Board of Directors of a Company. The probabilities of their winning are 0.5, 0.3, 0.2 respectively. If the group A wins, the probability of introducing a new product is 0.7 and the corresponding probabilities for group B and C are 0.6 and 0.5 respectively. Find the probability that the new product will be introduced.

Solution :

Given $P(A) = 0.5$, $P(B) = 0.3$ and $P(C) = 0.2$

$$\therefore P(A) + P(B) + P(C) = 1$$

then events A, B, C are exhaustive.

If $P(E) =$ Probability of introducing a new product, then as given

$$P(E|A) = 0.7, P(E|B) = 0.6 \text{ and } P(E|C) = 0.5$$

$$\therefore P(E) = P(A) \cdot P(E|A) + P(B) \cdot P(E|B) + P(C) \cdot P(E|C)$$

$$= 0.5 \times 0.7 + 0.3 \times 0.6 + 0.2 \times 0.5 = 0.35 + 0.18 + 0.10 = 0.63$$

Illustration 21 : A pair of dice is rolled together till a sum of either 5 or 7 is obtained. Find the probability that 5 comes before 7.

Solution : Let E_1 = the event of getting 5 in a roll of two dice = $\{(1, 4), (2, 3), (3, 2), (4, 1)\}$

$$\therefore P(E_1) = \frac{n(E_1)}{n(S)} = \frac{4}{6 \times 6} = \frac{1}{9}$$

Let E_2 = the event of getting either 5 or 7

$$= \{(1, 4), (2, 3), (3, 2), (4, 1), (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$$

$$\therefore P(E_2) = \frac{n(E_2)}{n(S)} = \frac{10}{6 \times 6} = \frac{5}{18}$$

$$\therefore \text{the probability of getting neither 5 nor 7} = P(\bar{E}_2) = 1 - P(E_2) = 1 - \frac{5}{18} = \frac{13}{18}$$

The event of getting 5 before 7 = $E_1 \cup (\bar{E}_2 E_1) \cup (\bar{E}_2 \bar{E}_2 E_1) \cup \dots$ to ∞

$\therefore$ the probability of getting 5 before 7

$$= P(E_1) + P(\bar{E}_2 E_1) + P(\bar{E}_2 \bar{E}_2 E_1) + \dots \text{ to } \infty = P(E_1) + P(\bar{E}_2)P(E_1) + P(\bar{E}_2)P(\bar{E}_2)P(E_1) + \dots \text{ to } \infty$$

$$= \frac{1}{9} + \frac{13}{18} \cdot \frac{1}{9} + \frac{13}{18} \cdot \frac{13}{18} \cdot \frac{1}{9} + \dots \text{ to } \infty$$

$$= \frac{1}{9} \left[1 + \frac{13}{18} + \left(\frac{13}{18} \right)^2 + \dots \text{ to } \infty \right] = \frac{1}{9} \cdot \frac{1}{1 - \frac{13}{18}} = \frac{1}{9} \cdot \frac{18}{5} = \frac{2}{5}$$

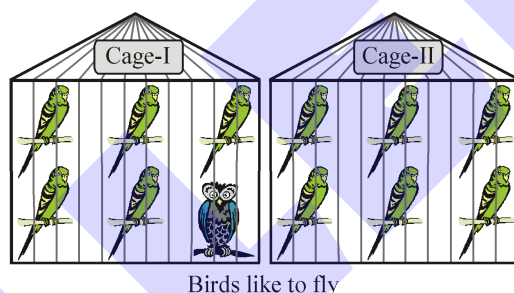
Do yourself - 6 :

1. An urn contains 6 white & 4 black balls. A die is rolled and the number of balls equal to the number obtained on the die are drawn from the urn. Find the probability that the balls drawn are all black.
2. There are n bags such that i^{th} bag ($1 \leq i \leq n$) contains i black and 2 white balls. Two balls are drawn from a randomly selected bag out of given n bags. Find the probability that the both drawn balls are white.
3. An unbiased die with numbers 1, 2, 3, 4, 6 and 8 on its six faces is rolled. After this roll if an odd number appears on the top face, all odd numbers on the die are doubled. If an even number appears on the top face, all the even numbers are halved. If the given die changes in this way then the probability that the face 2 will appear on the second roll is -
4. A butterfly randomly lands on one of the six squares of the T-shaped figure shown and then randomly moves to an adjacent square. The probability that the butterfly ends up on the R square is

- (A) $\frac{1}{4}$ (B) $\frac{1}{3}$
(C) $\frac{2}{3}$ (D) $\frac{1}{6}$

G			
Y	W	R	B
O			

5. Lot A consists of 3G and 2D articles. Lot B consists of 4G and 1D article. A new lot C is formed by taking 3 articles from A and 2 from B. The probability that an article chosen at random from C is defective, is
 (A) $1/3$ (B) $2/5$ (C) $8/25$ (D) none
6. On a normal standard die one of the 21 dots from any one of the six faces is removed at random with each dot equally likely to be chosen. The die is then rolled. The probability that the top face has an odd number of dots is
 (A) $\frac{5}{11}$ (B) $\frac{5}{12}$ (C) $\frac{11}{21}$ (D) $\frac{6}{11}$
7. A bowl has 6 red marbles and 3 green marbles. The probability that a blind folded person will draw a red marble on the second draw from the bowl without replacing the marble from the first draw, is
 (A) $2/3$ (B) $1/4$ (C) $5/12$ (D) $5/8$
8. Shalu bought two cages of birds : Cage-I contains 5 parrots and 1 owl, and Cage-II contains 6 parrots, as shown



One day Shalu forgot to lock both cages and two birds flew from Cage-I to Cage-II. Then two birds flew back from Cage-II to Cage-I. Assume that all birds have equal chance of flying, the probability that the Owl is still in Cage-I, is
 (A) $1/6$ (B) $1/3$ (C) $2/3$ (D) $3/4$

10. PROBABILITY OF THREE EVENTS :

For any three events A, B and C we have

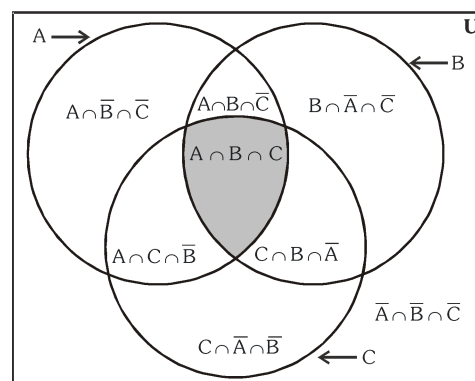
- (a) $P(\text{atleast one of A, B and C occurs})$

$$= P(A \text{ or } B \text{ or } C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

Note : $P(E_1 \cup E_2 \dots \cup E_n) = 1 - P(\bar{E}_1 \cap \bar{E}_2 \dots \cap \bar{E}_n)$

- (b) $P(\text{at least two of A,B,C occur})$

$$= P(B \cap C) + P(C \cap A) + P(A \cap B) - 2P(A \cap B \cap C)$$



- (c) $P(\text{exactly two of } A, B, C \text{ occur})$
 $= P(B \cap C) + P(C \cap A) + P(A \cap B) - 3P(A \cap B \cap C)$
- (d) $P(\text{exactly one of } A, B, C \text{ occurs})$
 $= P(A) + P(B) + P(C) - 2P(B \cap C) - 2P(C \cap A) - 2P(A \cap B) + 3P(A \cap B \cap C)$

Illustration 22 : Let A, B, C be three events. If the probability of occurring exactly one event out of A and B is $1 - a$, out of B and C is $1 - 2a$, out of C and A is $1 - a$ and that of occurring three events simultaneously is a^2 , then prove that the probability that at least one out of A, B, C will occur is greater than $1/2$.

Solution :

$$P(A) + P(B) - 2P(A \cap B) = 1 - a \quad \dots(1)$$

$$\text{and } P(B) + P(C) - 2P(B \cap C) = 1 - 2a \quad \dots(2)$$

$$\text{and } P(C) + P(A) - 2P(C \cap A) = 1 - a \quad \dots(3)$$

$$\text{and } P(A \cap B \cap C) = a^2 \quad \dots(4)$$

$$\therefore P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

$$= \frac{1}{2} \{ P(A) + P(B) - 2P(A \cap B) + P(B) + P(C) - 2P(B \cap C) + P(C) + P(A) - 2P(C \cap A) \} + P(A \cap B \cap C)$$

$$= \frac{1}{2} \{ 1 - a + 1 - 2a + 1 - a \} + a^2 \quad \{ \text{from (1), (2), (3) \& (4)} \}$$

$$= \frac{3}{2} - 2a + a^2 = (a - 1)^2 + \frac{1}{2} > \frac{1}{2}$$

Ans.

Do yourself - 7 :

1. In a class, there are 100 students out of which 45 study mathematics, 48 study physics, 40 study chemistry, 12 study both mathematics & physics, 11 study both physics & chemistry, 15 study both mathematics & chemistry and 5 study all three subjects. A student is selected at random, then find the probability that the selected student studies
- (a) only one subject (b) neither physics nor chemistry

11. BINOMIAL PROBABILITY DISTRIBUTION :

Suppose that we have an experiment such as tossing a coin or die repeatedly or choosing a marble from an urn repeatedly. Each toss or selection is called a trial. In any single trial there will be a probability associated with a particular event such as head on the coin, 4 on the die, or selection of a red marble. In some cases this probability will not change from one trial to the next (as in tossing a coin or die). Such trials are then said to be independent and are often called Bernoulli trials after James Bernoulli who investigated them at the end of the seventeenth century.

Let p be the probability that an event will happen in any single Bernoulli trial (called the probability of success). Then $q = 1 - p$ is the probability that the event will fail to happen in any single trial (called the probability of failure). The probability that the event will happen exactly x times in n trials (i.e., x successes and $n - x$ failures will occur) is given by the probability function.

$$f(x) = P(X = x) = \binom{n}{x} p^x q^{n-x} = \frac{n!}{x!(n-x)!} p^x q^{n-x} \quad \dots\dots\dots (i)$$

where the random variable X denotes the number of successes in n trials and $x = 0, 1, \dots, n$.

Example : The probability of getting exactly 2 heads in 6 tosses of a fair coin is

$$P(X = 2) = \binom{6}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{6-2} = \frac{6!}{2!4!} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{6-2} = \frac{15}{64}$$

The discrete probability function (i) is often called the binomial distribution since for $x = 0, 1, 2, \dots, n$, it corresponds to successive terms in the binomial expansion

$$(q + p)^n = q^n + \binom{n}{1} q^{n-1} p + \binom{n}{2} q^{n-2} p^2 + \dots + p^n = \sum_{x=0}^n \binom{n}{x} p^x q^{n-x}$$

Illustration 23 : If a fair coin is tossed 10 times, find the probability of getting

- (i) exactly six heads (ii) atleast six heads (iii) atmost six heads

Solution : The repeated tosses of a coin are Bernoulli trials. Let X denotes the number of heads in an experiment of 10 trials.

Clearly, X has the binomial distribution with $n = 10$ and $p = \frac{1}{2}$

Therefore $P(X = x) = {}^nC_x q^{n-x} p^x$, $x = 0, 1, 2, \dots, n$

Here $n = 10$, $p = \frac{1}{2}$, $q = 1 - p = \frac{1}{2}$

Therefore $P(X = x) = {}^{10}C_x \left(\frac{1}{2}\right)^{10-x} \left(\frac{1}{2}\right)^x = {}^{10}C_x \left(\frac{1}{2}\right)^{10}$

Now (i) $P(X = 6) = {}^{10}C_6 \left(\frac{1}{2}\right)^{10} = \frac{10!}{6! \times 4! 2^{10}} = \frac{105}{512}$

(ii) $P(\text{atleast six heads})$

$$= P(X \geq 6) = P(X = 6) + P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10)$$

$$= {}^{10}C_6 \left(\frac{1}{2}\right)^{10} + {}^{10}C_7 \left(\frac{1}{2}\right)^{10} + {}^{10}C_8 \left(\frac{1}{2}\right)^{10} + {}^{10}C_9 \left(\frac{1}{2}\right)^{10} + {}^{10}C_{10} \left(\frac{1}{2}\right)^{10}$$

$$= \left[\left(\frac{10!}{6! \times 4!}\right) + \left(\frac{10!}{7! \times 3!}\right) + \left(\frac{10!}{8! \times 2!}\right) + \left(\frac{10!}{9! \times 1!}\right) + \left(\frac{10!}{10!}\right) \right] \frac{1}{2^{10}} = \frac{193}{512}$$

$$\begin{aligned}
 \text{(iii) } P(\text{at most six heads}) &= P(X \leq 6) \\
 &= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6) \\
 &= \left(\frac{1}{2}\right)^{10} + {}^{10}C_1 \left(\frac{1}{2}\right)^{10} + {}^{10}C_2 \left(\frac{1}{2}\right)^{10} + {}^{10}C_3 \left(\frac{1}{2}\right)^{10} + {}^{10}C_4 \left(\frac{1}{2}\right)^{10} + {}^{10}C_5 \left(\frac{1}{2}\right)^{10} + {}^{10}C_6 \left(\frac{1}{2}\right)^{10} \\
 &= \frac{848}{1024} = \frac{53}{64}
 \end{aligned}$$

Illustration 24 : India and Pakistan play a 5 match test series of hockey, the probability that India wins at least three matches is -

- (A) $\frac{1}{2}$ (B) $\frac{3}{5}$ (C) $\frac{4}{5}$ (D) $\frac{5}{16}$

Solution : India win atleast three matches

$$= {}^5C_3 \left(\frac{1}{2}\right)^5 + {}^5C_4 \left(\frac{1}{2}\right)^5 + {}^5C_5 \left(\frac{1}{2}\right)^5 = \left(\frac{1}{2}\right)^5 (16) = \frac{1}{2} \quad \text{Ans. (A)}$$

Illustration 25 : A coin is tossed 7 times. Each time a man calls head. The probability that he wins the toss on more than three occasions is -

- (A) $\frac{1}{4}$ (B) $\frac{5}{8}$ (C) $\frac{1}{2}$ (D) none of these

Solution : The man has to win at least 4 times.

$\therefore$ the required probability

$$\begin{aligned}
 &= {}^7C_4 \left(\frac{1}{2}\right)^4 \cdot \left(\frac{1}{2}\right)^3 + {}^7C_5 \cdot \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^2 + {}^7C_6 \left(\frac{1}{2}\right)^6 \cdot \frac{1}{2} + {}^7C_7 \left(\frac{1}{2}\right)^7 \\
 &= ({}^7C_4 + {}^7C_5 + {}^7C_6 + {}^7C_7) \cdot \frac{1}{2^7} = \frac{64}{2^7} = \frac{1}{2}. \quad \text{Ans. (C)}
 \end{aligned}$$

Do yourself - 8 :

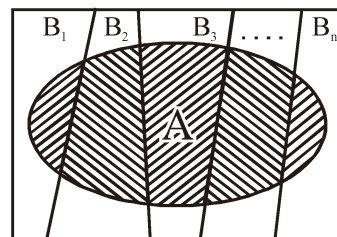
1. An experiment succeeds twice as often as it fails. Find the probability that in next 6 trials, there will be more than 3 successes.
2. Find the probability of getting 4 exactly thrice in 7 throws of a die.
3. An examination consists of 8 questions in each of which one of the 5 alternatives is the correct one. On the assumption that a candidate who has done no preparatory work chooses for each question any one of the five alternatives with equal probability, the probability that he gets more than one correct answer is equal to

- (A) $(0.8)^8$ (B) $3 (0.8)^8$ (C) $1 - (0.8)^8$ (D) $1 - 3 (0.8)^8$

12. BAYE'S THEOREM :

Let an event A of an experiment occurs with its n mutually exclusive & exhaustive events $B_1, B_2, B_3, \dots, B_n$ & the probabilities $P(A/B_1), P(A/B_2), \dots, P(A/B_n)$ are known, then

$$P(B_i/A) = \frac{P(B_i) \cdot P(A/B_i)}{\sum_{i=1}^n P(B_i) \cdot P(A/B_i)}$$



Explanation :

$A \equiv$ event what we have ; $B_i \equiv$ event what we want & remaining are alternative events.

Now, $P(AB_i) = P(A) \cdot P(B_i/A) = P(B_i) \cdot P(A/B_i)$

$$P(B_i/A) = \frac{P(B_i) \cdot P(A/B_i)}{P(A)} = \frac{P(B_i) \cdot P(A/B_i)}{\sum_{i=1}^n P(AB_i)}$$

$$P(B_i/A) = \frac{P(B_i) \cdot P(A/B_i)}{\sum_{i=1}^n P(B_i) \cdot P(A/B_i)}$$

Illustration 26 : Given three identical boxes I, II and III, each containing two coins. In box I, both coins are gold coins, in box II, both are silver coins and in the box III, there is one gold and one silver coin. A person chooses a box at random and takes out a coin. If the coin is of gold, what is the probability that the other coin in the box is also of gold ?

Solution : Let E_1, E_2 and E_3 be the events that boxes I, II and III are chosen, respectively.

$$\text{Then } P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$$

Also, let A be the event that 'the coin drawn is of gold'

$$\text{Then } P(A|E_1) = P(\text{a gold coin from box I}) = \frac{2}{2} = 1$$

$$P(A|E_2) = P(\text{a gold coin from box II}) = 0$$

$$P(A|E_3) = P(\text{a gold coin from box III}) = \frac{1}{2}$$

Now, the probability that the other coin in the box is of gold

= the probability that gold coin is drawn from the box I.

$$= P(E_1|A)$$

By Baye's theorem, we know that

$$P(E_1|A) = \frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2) + P(E_3)P(A|E_3)} = \frac{\frac{1}{3} \times 1}{\frac{1}{3} \times 1 + \frac{1}{3} \times 0 + \frac{1}{3} \times \frac{1}{2}} = \frac{2}{3}$$

Illustration 27 : A bag A contains 2 white and 3 red balls and a bag B contains 4 white and 5 red balls. One ball is drawn at random from one of the bags and it is found to be red. Find the probability that it was drawn from the bag B.

Solution : Let E_1 = The event of ball being drawn from bag A

E_2 = The event of ball being drawn from bag B.

E = The event of ball being red.

Since, both the bags are equally likely to be selected, therefore

$$P(E_1) = P(E_2) = \frac{1}{2} \text{ and } P(E|E_1) = \frac{3}{5} \text{ and } P(E|E_2) = \frac{5}{9}$$

$\therefore$ Required probability

$$P(E_2|E) = \frac{P(E_2)P(E|E_2)}{P(E_1) \cdot P(E|E_1) + P(E_2)P(E|E_2)} = \frac{\frac{1}{2} \times \frac{5}{9}}{\frac{1}{2} \times \frac{3}{5} + \frac{1}{2} \times \frac{5}{9}} = \frac{25}{52}$$

Do yourself - 9 :

1. A pack of cards was found to contain only 51 cards. If first 13 cards, which are examined, are all red, then find the probability that the missing card is black.
2. A man has 3 coins A, B & C. A is fair coin. B is biased such that the probability of occurring head on it is $\frac{2}{3}$. C is also biased with the probability of occurring head as $\frac{1}{3}$. If one coin is selected and tossed three times, giving two heads and one tail, find the probability that the chosen coin was A.
3. An instrument consists of two units. Each unit must function for the instrument to operate. The reliability of the first unit is 0.9 & that of the second unit is 0.8. The instrument is tested & fails. The probability that "only the first unit failed & the second unit is sound" is :
(A) $\frac{1}{7}$ (B) $\frac{2}{7}$ (C) $\frac{3}{7}$ (D) $\frac{4}{7}$
4. A purse contains 2 six sided dice. One is a normal fair die, while the other has 2 ones, 2 threes, and 2 fives. A die is picked up and rolled. Because of some secret magnetic attraction of the unfair die, there is 75% chance of picking the unfair die and a 25% chance of picking a fair die. The die is rolled and shows up the face 3. The probability that a fair die was picked up, is
(A) $\frac{1}{7}$ (B) $\frac{1}{4}$ (C) $\frac{1}{6}$ (D) $\frac{1}{24}$
5. On a Saturday night 20% of all drivers in U.S.A. are under the influence of alcohol. The probability that a driver under the influence of alcohol will have an accident is 0.001. The probability that a sober driver will have an accident is 0.0001. If a car on a Saturday night smashed into a tree, the probability that the driver was under the influence of alcohol, is -
(A) $\frac{3}{7}$ (B) $\frac{4}{7}$ (C) $\frac{5}{7}$ (D) $\frac{6}{7}$
6. A box has four dice in it. Three of them are fair dice but the fourth one has the number five on all of its faces. A die is chosen at random from the box and is rolled three times and shows up the face five on all the three occasions. The chance that the die chosen was a rigged die, is
(A) $\frac{216}{217}$ (B) $\frac{215}{219}$ (C) $\frac{216}{219}$ (D) none

7. In a certain factory, machines A, B and C produce bolts. Of their production, machines A, B, and C produce 2%, 1% and 3% defective bolts respectively. Machine A produces 35% of the total output of bolts, machine B produces 25% and machine C produces 40%. A bolt is chosen at random from the factory's production and is found to be defective. The probability it was produced on machine C, is
- (A) $\frac{6}{11}$ (B) $\frac{23}{45}$ (C) $\frac{24}{43}$ (D) $\frac{3}{11}$
8. A purse contains 100 coins of unknown value, a coin drawn at random is found to be a rupee. The chance that it is the only rupee in the purse, is (Assume all numbers of rupee coins in the purse to be equally likely.)
- (A) $\frac{1}{5050}$ (B) $\frac{2}{5151}$ (C) $\frac{1}{4950}$ (D) $\frac{2}{4950}$
9. Mr. Dupont is a professional wine taster. When given a French wine, he will identify it with probability 0.9 correctly as French, and will mistake it for a Californian wine with probability 0.1. When given a Californian wine, he will identify it with probability 0.8 correctly as Californian, and will mistake it for a French wine with probability 0.2. Suppose that Mr. Dupont is given ten unlabelled glasses of wine, three with French and seven with Californian wines. He randomly picks a glass, tries the wine, and solemnly says : "French". The probability that the wine he tasted was Californian, is nearly equal to
- (A) 0.14 (B) 0.24 (C) 0.34 (D) 0.44

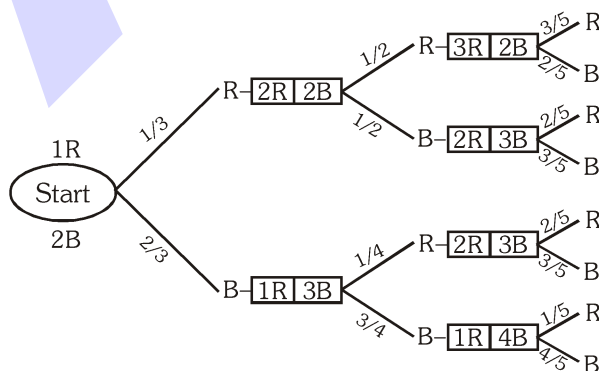
13. PROBABILITY THROUGH STATISTICAL (STOCHASTIC) TREE DIAGRAM :

These tree diagrams are generally drawn by economist and give a simple approach to solve a problem.

Illustration 28 : A bag initially contains 1 red ball and 2 blue balls. A trial consists of selecting a ball at random, noting its colour and replacing it together with an additional ball of the same colour. Given that three trials are made, draw a tree diagram illustrating the various probabilities. Hence, or otherwise, find the probability that

- (a) at least one blue ball is drawn
 (b) exactly one blue ball is drawn
 (c) Given that all three balls drawn are of the same colour find the probability that they are all red.

Solution :



Calculations :

$$P(A) = 1 - P(RRR) = 1 - \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{3}{5} = 1 - \frac{1}{10} = \frac{9}{10}$$

$$P(\text{exactly one Blue}) = \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{2}{5} + \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{2}{5} + \frac{2}{3} \cdot \frac{1}{4} \cdot \frac{2}{5} = \frac{1}{15} + \frac{1}{15} + \frac{1}{15} = \frac{3}{15} = \frac{1}{5}$$

$$P(C) = P\left(\frac{RRR}{(RRR \cup BBB)}\right) = \frac{P(RRR)}{P(RRR) + P(BBB)} = \frac{\frac{1}{3} \cdot \frac{1}{2} \cdot \frac{3}{5}}{\frac{1}{3} \cdot \frac{1}{2} \cdot \frac{3}{5} + \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5}} = \frac{\frac{1}{10}}{\frac{1}{10} + \frac{4}{10}} = \frac{1}{5}$$

Do yourself - 10 :

1. If at least one child in a family with 3 children is a boy then the probability that exactly 2 of the children are boys, is

(A) 3/7 (B) 4/7 (C) 1/3 (D) 3/8

2. From an urn containing six balls, 3 white and 3 black ones, a person selects at random an even number of balls (all the different ways of drawing an even number of balls are considered equally probable, irrespective of their number). Then the probability that there will be the same number of black and white balls among them

(A) 4/5 (B) 11/15 (C) 11/30 (D) 2/5

3. There are three main political parties namely 1, 2, 3. If in the adjoining table p_{ij} , ($i, j=1, 2, 3$) denote the probability that party j wins the general elections contested when party i is in the power. What is the probability that the party 2 will be in power after the next two elections, given that the party 1 is in the power?

$P_{11}=0.7$	$P_{12}=0.2$	$P_{13}=0.1$
$P_{21}=0.5$	$P_{22}=0.3$	$P_{23}=0.2$
$P_{31}=0.3$	$P_{32}=0.4$	$P_{33}=0.3$

(A) 0.27 (B) 0.24 (C) 0.14 (D) 0.06

4. Suppose families always have one, two or three children, with probabilities $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{1}{4}$ respectively. Assume everyone eventually gets married and has children, the probability of a couple having exactly four grandchildren is

(A) $\frac{27}{128}$ (B) $\frac{37}{128}$ (C) $\frac{25}{128}$ (D) $\frac{20}{128}$

5. Miss C has either Tea or Coffee at morning break. If she has tea one morning, the probability she has tea the next morning is 0.4. If she has coffee one morning, the probability she has coffee next morning is 0.3. Suppose she has coffee on a Monday morning. The probability that she has tea on the following Wednesday morning is

(A) 0.46 (B) 0.49 (C) 0.51 (D) 0.61

6. One bag contains 3 white & 2 black balls, and another contains 2 white & 3 black balls. A ball is drawn from the second bag & placed in the first, then a ball is drawn from the first bag & placed in the second. When the pair of the operations is repeated, the probability that the first bag will contain 5 white balls is:

(A) 1/25 (B) 1/125 (C) 1/225 (D) 2/15

Miscellaneous Illustrations :

Illustration 29 : Three persons A, B, C in order cut a pack of cards, replacing them after each cut, on the condition that the first who cuts a spade shall win a prize; find their respective chances.

Solution : Let p be the chance of cutting a spade and q the chance of not cutting a spade from a pack of 52 cards.

$$\text{Then } p = \frac{{}^{13}C_1}{{}^{52}C_1} = \frac{1}{4} \text{ and } q = 1 - \frac{1}{4} = \frac{3}{4}$$

Now A will win a prize if he cuts spade at 1st, 4th, 7th, 10th turns, etc. Note that A will get a second chance if A, B, C all fail to cut a spade once and then A cuts a spade at the 4th turn. Similarly he will cut a spade at the 7th turn when A, B, C fail to cut spade twice, etc.

$$\text{Hence A's chance of winning the prize} = p + q^3p + q^6p + q^9p + \dots = \frac{p}{1-q^3} = \frac{\frac{1}{4}}{1-\left(\frac{3}{4}\right)^3} = \frac{16}{37}$$

$$\text{Similarly B's chance} = (qp + q^4p + q^7p + \dots) = q(p + q^3p + q^6p + \dots) = \frac{3}{4} \cdot \frac{16}{37} = \frac{12}{37}$$

$$\text{and C's chance} = \frac{3}{4} \text{ of B's chance} = \frac{3}{4} \cdot \frac{12}{37} = \frac{9}{37}$$

Illustration 30 : (a) If p and q are chosen randomly from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, with replacement, determine the probability that the roots of the equation $x^2 + px + q = 0$ are real.

(b) Each coefficient in the equation $ax^2 + bx + c = 0$ is determined by throwing an ordinary die. Find the probability that the equation will have equal roots.

Solution : (a) If roots of $x^2 + px + q = 0$ are real, then $p^2 - 4q \geq 0$ (i)
Both p, q belongs to set $S = \{1, 2, 3, \dots, 10\}$ when $p = 1$, no value of q from S will satisfy (i)

$p = 2$ $q = 1$ will satisfy 1 value

$p = 3$ $q = 1, 2$ 2 value

$p = 4$ $q = 1, 2, 3, 4$ 4 value

$p = 5$ $q = 1, 2, 3, 4, 5, 6$ 6 value

$p = 6$ $q = 1, 2, 3, 4, 5, 6, 7, 8, 9$ 9 value

For $p = 7, 8, 9, 10$ all the ten values of q will satisfy.

Sum of these selections is $1 + 2 + 4 + 6 + 9 + 10 + 10 + 10 + 10 = 62$

But the total number of selections of p and q without any order is $10 \times 10 = 100$

$$\text{Hence the required probability is} = \frac{62}{100} = 0.62$$

(b) Roots equal $\Rightarrow b^2 - 4ac = 0$

$$\therefore \left(\frac{b}{2}\right)^2 = ac \quad \dots (i)$$

Each coefficient is an integer, so we consider the following cases :

$$b = 1 \quad \therefore \quad \frac{1}{4} = ac$$

No integral values of a and c

$$b = 2 \quad 1 = ac \quad \therefore \quad (1, 1)$$

$$b = 3 \quad 9/2 = ac$$

No integral values of a and c

$$b = 4 \quad 4 = ac \quad \therefore \quad (1, 4), (2, 2), (4, 1)$$

$$b = 5 \quad 25/2 = ac$$

No integral values of a and c

$$b = 6 \quad 9 = ac \quad \therefore \quad (3, 3)$$

Thus we have 5 favourable ways for $b = 2, 4, 6$

Total number of equations is $6 \cdot 6 \cdot 6 = 216$

$$\therefore \quad \text{Required probability is } \frac{5}{216}$$

Illustration 31: A set A has n elements. A subset P of A is selected at random. Returning the element of P, the set A is formed again and then a subset Q is selected from it. Find the probability that P and Q have no common elements.

Solution : Let P be the empty set, or one element set or two elements set or n elements set. Then the set Q will be chosen from amongst the remaining n elements or $n - 1$ elements or $n - 2$ elements or no elements. The probability of P being an empty set is ${}^nC_0/2^n$, the probability of P being one element set is ${}^nC_1/2^n$ and in general, the probability of P being an r element set is ${}^nC_r/2^n$.

When the set P consisting of r elements is chosen from A, then the probability of choosing the set Q from amongst the remaining $n - r$ elements is $2^{n-r}/2^n$. Hence the probability that P and Q have no common elements is given by

$$\sum_{r=0}^n \frac{{}^nC_r}{2^n} \cdot \frac{2^{n-r}}{2^n} = \frac{1}{4^n} \sum_{r=0}^n {}^nC_r \cdot 2^{n-r} = \left(\frac{1}{4}\right)^n (1+2)^n = \left(\frac{3}{4}\right)^n \quad [\text{By binomial theorem}]$$

Illustration 32 : The probabilities of three events A, B and C are $P(A) = 0.6$, $P(B) = 0.4$ and $P(C) = 0.5$. If $P(A \cup B) = 0.8$, $P(A \cap C) = 0.3$, $P(A \cap B \cap C) = 0.2$ and $P(A \cup B \cup C) \geq 0.85$, find $P(B \cap C)$.

Solution : $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$0.8 = 0.6 + 0.4 - P(A \cap B)$$

$$\therefore \quad P(A \cap B) = 0.2$$

$$\begin{aligned} \text{Now } P(A \cup B \cup C) &= S_1 - S_2 + S_3 = (0.6 + 0.4 + 0.5) - (0.2 + P(B \cap C) + 0.3) + 0.2 \\ &= 1.5 - 0.3 - P(B \cap C) \end{aligned}$$

We know $0.85 \leq P(A \cup B \cup C) \leq 1$

$$\text{or } 0.85 \leq 1.2 - P(B \cap C) \leq 1$$

$$\therefore \quad 0.2 \leq P(B \cap C) \leq 0.35$$